

DevOps Documentation

Program 7: Deploy a Web Application to Kubernetes using Minikube

Project-7

—> nginx-deployment.yaml

S1. Start minikube

minikube start

```
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program7$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
📡 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image docker.io/kubernetes/dashboard:v2.7.0
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ▪ Using image docker.io/kubernetes/metrics-server:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌞 Enabled addons: storage-provisioner, dashboard, default-storageclass
📢 Done! kubectrl is now configured to use "minikube" cluster and "default" namespace by default
```

S2. Create a Deployment file nginx-deployment.yaml

```
! nginx-deployment.yaml x
! nginx-deployment.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: nginx-deployment
5  spec:
6    replicas: 3
7    selector:
8      matchLabels:
9        app: nginx
10   template:
11     metadata:
12       labels:
13         app: nginx
14     spec:
15       containers:
16       - name: nginx
17         image: nginx:latest
18         ports:
19         - containerPort: 80
```

S3. Apply the nginx-deployment.yaml

kubectl apply -f nginx-deployment.yaml

```
1rv24mc113_varun@varun-srivastava:~/DevOpsLab/program7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment configured
1rv24mc113_varun@varun-srivastava:~/DevOpsLab/program7$
```

S4. Check status

kubectl get deployments

kubectl get pods

```
1rv24mc113_varun@varun-srivastava:~/DevOpsLab/program7$ kubectl get deployments
kubectl get pods
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    3/3     3            3           5d23h
NAME                READY   STATUS    RESTARTS   AGE
my-nginx-pod        1/1     Running   1 (27h ago)  28h
nginx-deployment-6f9664446b-dkwvc  1/1     Running   0          53s
nginx-deployment-6f9664446b-v529x  1/1     Running   1 (27h ago)  28h
nginx-deployment-6f9664446b-w7rls  1/1     Running   1 (27h ago)  28h
1rv24mc113_varun@varun-srivastava:~/DevOpsLab/program7$
```

S5. Expose the Deployment as service

kubectl expose deployment nginx-deployment --type=NodePort --port=80

```
1rv24mc113_varun@varun-srivastava:~/DevOpsLab/program7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:32238

🖼️ Opening service default/nginx-deployment in default browser...

```
1rv24mc113_varun@varun-srivastava:~/DevOpsLab/program7$ Opening in existing browser session.
```

S6. Access the Web App

minikube service nginx-deployment

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.